

MODULE:1

Matrix Methods to Linear Differential Equations.

Matrix

A matrix is a rectangular array of numbers arranged in ' m ' horizontal rows and ' n ' vertical columns.

The numbers are called entries of the matrix and such a matrix is said to be dimension $m \times n$ (denoted by m by n).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Here A is a matrix of order mn . It has m rows and n columns. Each of mn numbers is called an element of the matrix.

Eigen Values and Eigen Vectors

Let A be the given square matrix. If there exist a non-zero column vector X and a scalar λ such that $AX = \lambda X$, then λ is called an Eigen value and X is called an Eigen vector corresponding to the Eigen values λ .

Characteristic Equation

The equation $|A - \lambda I| = 0$ is called the characteristic equation of the matrix A .

Eigen Value

Solving the characteristic equation $|A - \lambda I| = 0$ we get the values of λ and these values are called Eigen values or Latent roots or Characteristic roots.

Eigen Vector

The vector $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ satisfying the equation $(A - \lambda I)X = 0$ is called an Eigen vector.

Working rule to find the Characteristic Equation

If A is a square matrix of order 3, then its characteristic equation is

$$|A - \lambda I| = 0$$

$$\text{i.e., } \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = 0$$

Expand the determinant and we get the characteristic equation (cubic equation)

(Or)

The characteristic equation can be written as $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

S_2 = sum of the minors of principal diagonal elements

S_3 = determinant of A (or) $|A|$

Note

If A is a square matrix of order 2, then its characteristic equation can be written as $\lambda^2 - S_1\lambda + S_2 = 0$

Where S_1 = Sum of main diagonal elements

S_2 = Determinant of A or $|A|$

Illustrative Examples

Example 1 : Find the eigenvalues and eigenvectors of the matrix $A =$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

Solution : Given $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 1 + 5 + 1 = 7$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix}$$

$$= (5 - 1) + (1 - 9) + (5 - 1) = 4 - 8 + 4 = 0$$

$$S_3 = |A| = \begin{vmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{vmatrix}$$

$$= 1(5 - 1) - 1(1 - 3) + 3(1 - 15)$$

$$= 1(4) - 1(-2) + 3(-14)$$

$$= 4 + 2 - 42 = -36$$

$\therefore$ The characteristic equation is $\lambda^3 - 7\lambda^2 + 0\lambda + 36 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} -2 & 1 & -7 & 0 & 36 \\ & & 0 & -2 & 18 & -36 \\ \hline & 1 & -9 & 18 & 0 \end{array}$$

i. e., $\lambda = -2, \lambda^2 - 9\lambda + 18 = 0$

i. e., $\lambda = -2, (\lambda - 3)(\lambda - 6) = 0$

$\therefore$ The eigenvalues are $\lambda = -2, 3, 6$

To find the eigenvector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} 1 - \lambda & 1 & 3 \\ 1 & 5 - \lambda & 1 \\ 3 & 1 & 1 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (1 - \lambda)x_1 + 1x_2 + 3x_3 &= 0 \\ 1x_1 + (5 - \lambda)x_2 + 1x_3 &= 0 \\ 3x_1 + 1x_2 + (1 - \lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = -2$

(1) $\Rightarrow$ becomes, $3x_1 + 1x_2 + 3x_3 = 0$

$$1x_1 + 7x_2 + 1x_3 = 0$$

Solving by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 3 & 3 \\ \swarrow & \searrow & \swarrow \\ 7 & 1 & 1 \end{array} \quad \begin{array}{ccc} 3 & 3 & 1 \\ \swarrow & \searrow & \swarrow \\ 1 & 1 & 7 \end{array}$$

$$\frac{x_1}{1 - 21} = \frac{x_2}{3 - 3} = \frac{x_3}{21 - 1}$$

$$\frac{x_1}{-20} = \frac{x_2}{0} = \frac{x_3}{20} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{0} = \frac{x_3}{1}$$

$$X_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Case (ii) : $\lambda = 3$

(1) $\Rightarrow$ becomes, $-2x_1 + 1x_2 + 3x_3 = 0$

$$1x_1 + 2x_2 + 1x_3 = 0$$

Solving by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 3 & -2 \\ 2 & 1 & 1 \end{array} \quad \begin{array}{ccc} -2 & 1 & 1 \\ 1 & 1 & 2 \end{array}$$

$$\frac{x_1}{1-6} = \frac{x_2}{3+2} = \frac{x_3}{-4-1}$$

$$\frac{x_1}{-5} = \frac{x_2}{5} = \frac{x_3}{-5} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} = \frac{x_3}{-1}$$

$$X_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

Case (iii) : $\lambda = 6$

$$(1) \Rightarrow \text{becomes, } -5x_1 + 1x_2 + 3x_3 = 0$$

$$1x_1 - x_2 + 1x_3 = 0$$

Solving by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 3 & -5 \\ -1 & 1 & 1 \end{array} \quad \begin{array}{ccc} -5 & 1 & 1 \\ 1 & 1 & 2 \end{array}$$

$$\frac{x_1}{1+3} = \frac{x_2}{3+5} = \frac{x_3}{5-1}$$

$$\frac{x_1}{4} = \frac{x_2}{8} = \frac{x_3}{4} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{1}$$

$$X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Therefore, the eigen vectors are $X_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, X_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

Example 2 : Find the eigenvalues and eigenvectors of the matrix $A =$

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Solution : Given $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 0 + 0 + 0 = 0$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$= (0-1) + (0-1) + (0-1) = -1-1-1 = -3$$

$S_3 = |A|$

$$= \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= 0(0-1) - 1(0-1) + 1(1-0)$$

$$= 1 + 1 = 2$$

$\therefore$ The characteristic equation is $\lambda^3 - 0\lambda^2 - 3\lambda - 2 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -3 & -2 \\ & & -1 & 1 & 2 \\ \hline & 1 & -1 & -2 & 0 \end{array}$$

$$\text{i.e., } \lambda = -1, \lambda^2 - \lambda - 2 = 0$$

$$\text{i.e., } \lambda = -1, (\lambda + 1)(\lambda - 2) = 0$$

$\therefore$ The eigenvalues are $\lambda = -1, -1, 2$

To find the eigenvector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} -\lambda x_1 + 1x_2 + 1x_3 &= 0 \\ 1x_1 - \lambda x_2 + 1x_3 &= 0 \\ 1x_1 + 1x_2 - \lambda x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = -1$

(1) $\Rightarrow$ becomes, $x_1 + x_2 + x_3 = 0$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

All the three equations reduce to one and the same equation $x_1 + x_2 + x_3 = 0$.

Therefore, take any one equation and solve. i.e., $x_1 + x_2 + x_3 = 0$.

Let $x_3 = 0$

$$\begin{aligned} \therefore x_1 + x_2 &= 0 \Rightarrow x_1 = -x_2 \Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} \\ &\Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} = \frac{x_3}{0} \end{aligned}$$

$$X_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

Similarly, for **Case (ii)**

Let $x_2 = 0$

$$\begin{aligned} \therefore x_1 + x_3 &= 0 \Rightarrow x_1 = -x_3 \Rightarrow \frac{x_1}{-1} = \frac{x_3}{1} \\ &\Rightarrow \frac{x_1}{-1} = \frac{x_2}{0} = \frac{x_3}{1} \end{aligned}$$

$$X_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Case (iii) : $\lambda = 2$

(1) $\Rightarrow$ becomes, $-2x_1 + 1x_2 + 1x_3 = 0$

$$1x_1 - 2x_2 + 1x_3 = 0$$

Solving by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 1 & -2 \\ -2 & 1 & 1 \end{array} \quad \begin{array}{ccc} 1 & -2 & 1 \\ 1 & 1 & -2 \end{array} \quad \begin{array}{ccc} -2 & 1 & 1 \\ 1 & -2 & 1 \end{array}$$

$$\frac{x_1}{1+2} = \frac{x_2}{1+2} = \frac{x_3}{4-1}$$

$$\frac{x_1}{3} = \frac{x_2}{3} = \frac{x_3}{3} \Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1}$$

$$X_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Therefore, the eigen vectors are $X_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, X_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, X_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Note : Though two of the eigenvalues are equal, the eigen vectors X_1, X_2, X_3 are found to be linearly independent.

Example 3 : Find the eigenvalues and eigenvectors of the following matrix

$$A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$$

Solution : Given $A = \begin{pmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 2 + 1 - 1 = 2$$

S_2 = sum of the minors of principal diagonal elements

$$\begin{aligned}
 &= \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix} \\
 &= (-1 - 3) + (-2 - 2) + (2 + 2) \\
 &= -4 - 4 + 4 \\
 &= -4
 \end{aligned}$$

$$S_3 = |A|$$

$$\begin{aligned}
 &= \begin{vmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{vmatrix} \\
 &= 2(-1 - 3) + 2(-1 - 1) + 2(3 - 1) \\
 &= -8 - 4 + 4 = -8
 \end{aligned}$$

$\therefore$ The characteristic equation is $\lambda^3 - 2\lambda^2 - 4\lambda + 8 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr}
 2 & 1 & -2 & -4 & 8 \\
 & & 2 & 0 & -8 \\
 \hline
 & 1 & 0 & -4 & 0
 \end{array}$$

$$\text{i.e., } \lambda = 2, \lambda^2 - 4 = 0$$

$$\text{i.e., } \lambda = 2, \quad (\lambda + 2)(\lambda - 2) = 0$$

$\therefore$ The eigenvalues are $\lambda = 2, 2, -2$

To find the eigenvector

$$\text{Let } (A - \lambda I)X = 0, \text{ where } X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{aligned}
 &\begin{pmatrix} 2 - \lambda & -2 & 2 \\ 1 & 1 - \lambda & 1 \\ 1 & 3 & -1 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\left. \begin{aligned} (2 - \lambda)x_1 - 2x_2 + 2x_3 &= 0 \\ 1x_1 + (1 - \lambda)x_2 + 1x_3 &= 0 \\ 1x_1 + 3x_2 + (-1 - \lambda)x_3 &= 0 \end{aligned} \right\} \dots (1)
 \end{aligned}$$

Case (i) : $\lambda = 2$

$$(1) \Rightarrow \text{becomes, } 0x_1 - 2x_2 + 2x_3 = 0$$

$$1x_1 - 1x_2 + 1x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc}
 -2 & \nearrow & 2 \\
 & \searrow & \nearrow \\
 -1 & & 1
 \end{array}
 \begin{array}{ccc}
 2 & \nearrow & 0 \\
 & \searrow & \nearrow \\
 1 & & 1
 \end{array}
 \begin{array}{ccc}
 0 & \nearrow & -2 \\
 & \searrow & \nearrow \\
 1 & & -1
 \end{array}$$

$$\frac{x_1}{-2+2} = \frac{x_2}{2-0} = \frac{x_3}{0+2}$$

$$\frac{x_1}{0} = \frac{x_2}{2} = \frac{x_3}{2} \Rightarrow \frac{x_1}{0} = \frac{x_2}{1} = \frac{x_3}{1}$$

$$X_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{Also, } X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Case (ii) : $\lambda = -2$

$$(1) \Rightarrow \text{becomes, } 4x_1 - 2x_2 + 2x_3 = 0$$

$$1x_1 + 3x_2 + 1x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc}
 -2 & \nearrow & 2 \\
 & \searrow & \nearrow \\
 3 & & 1
 \end{array}
 \begin{array}{ccc}
 2 & \nearrow & 4 \\
 & \searrow & \nearrow \\
 1 & & 1
 \end{array}
 \begin{array}{ccc}
 4 & \nearrow & -2 \\
 & \searrow & \nearrow \\
 1 & & 3
 \end{array}$$

$$\frac{x_1}{-2-6} = \frac{x_2}{2-4} = \frac{x_3}{12+2}$$

$$\frac{x_1}{-8} = \frac{x_2}{-2} = \frac{x_3}{14} \Rightarrow \frac{x_1}{-4} = \frac{x_2}{-1} = \frac{x_3}{7}$$

$$X_3 = \begin{pmatrix} -4 \\ -1 \\ 7 \end{pmatrix}$$

Therefore, the eigen vectors are $X_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, X_3 = \begin{pmatrix} -4 \\ -1 \\ 7 \end{pmatrix}$

Note : Two eigenvalues are equal and the eigenvectors are linearly dependent.

Example 4 : Find the eigenvalues and eigenvectors of the matrix $A =$

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Solution : Given $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 2 + 2 + 2 = 6$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = (4 - 0) + (4 - 0) + (4 - 0) = 12$$

$$S_3 = |A|$$

$$= \begin{vmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= 2(4 - 0) - 1(0 - 0) + 0(0 - 0) = 8$$

$\therefore$ The characteristic equation is $\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} 2 & 1 & -6 & 12 & -8 \\ & & 2 & -8 & 8 \\ \hline & 1 & -4 & 4 & 0 \end{array}$$

$$i.e., \quad \lambda = 2, \lambda^2 - 4\lambda + 4 = 0$$

$$i.e., \quad \lambda = 2, (\lambda - 2)(\lambda - 2) = 0$$

$\therefore$ The Eigen Values are $\lambda = 2, 2, 2$

To find the eigen vector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} 2 - \lambda & 1 & 0 \\ 0 & 2 - \lambda & 1 \\ 0 & 0 & 2 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{array}{l} (2 - \lambda)x_1 + 1.x_2 + 0.x_3 = 0 \\ 0.x_1 + (2 - \lambda)x_2 + 1.x_3 = 0 \\ 0.x_1 + 0.x_2 + (2 - \lambda).x_3 = 0 \end{array} \right\} \dots (1)$$

Case (i) : $\lambda = 2$

$$(1) \Rightarrow \text{becomes, } 0.x_1 + 1.x_2 + 0.x_3 = 0$$

$$0.x_1 + 0.x_2 + 1.x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & \nearrow & 0 \\ 0 & \searrow & 1 \end{array} \quad \begin{array}{ccc} 0 & \nearrow & 0 \\ 0 & \searrow & 1 \end{array} \quad \begin{array}{ccc} 0 & \nearrow & 1 \\ 0 & \searrow & 0 \end{array}$$

$$\frac{x_1}{1 - 0} = \frac{x_2}{0 - 0} = \frac{x_3}{0 - 0} \Rightarrow \frac{x_1}{1} = \frac{x_2}{0} = \frac{x_3}{0}$$

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ Corresponding to the eigen value } \lambda = 2.$$

$\therefore$ The only eigen vector is $X = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ corresponding to the eigen values $\lambda =$

2.

Example 5 : Find the eigen values and eigen vectors of $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

Solution : Given $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

The characteristic equation is $\lambda^2 - S_1\lambda + S_2 = 0$

S_1 = sum of the principal diagonal elements

$$= 4 + 2 = 6$$

$S_2 = \det A = |A|$

$$= \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} = 8 - 3 = 5$$

$\therefore$ The characteristic equation is $\lambda^2 - 6\lambda + 5 = 0$

$$\text{i.e., } (\lambda - 1)(\lambda - 5) = 0$$

$\therefore$ The Eigen Values are $\lambda = 1, 5$

To find the Eigen Vector

Let $(A - \lambda I)X = 0$ where $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} 4 - \lambda & 1 \\ 3 & 2 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} (4 - \lambda)x_1 + 1x_2 = 0 \\ 3x_1 + (2 - \lambda)x_2 = 0 \end{cases} \text{ --- (1)}$$

Case (i) : $\lambda = 5$

$$(1) \Rightarrow \text{becomes, } -x_1 + 1x_2 = 0$$

$$3x_1 - 3x_2 = 0$$

$$\text{Let } -x_1 + 1x_2 = 0 \Rightarrow -x_1 = -x_2$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{1}$$

$$X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Case (ii) : $\lambda = 1$

$$(1) \Rightarrow \text{becomes, } 3x_1 + 1x_2 = 0$$

$$3x_1 + 1x_2 = 0$$

$$\text{Let } 3x_1 + 1x_2 = 0 \Rightarrow 3x_1 = -x_2 \Rightarrow \frac{x_1}{-1} = \frac{x_2}{3}$$

$$X_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

$\therefore$ The eigen vectors are $X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $X_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

Properties of Eigen Values

Property : 1 A square matrix A and its transpose A^T have the same eigen values.

Property : 2 The sum of the eigen values of a matrix A is equal to the sum of the principal diagonal elements of A . (i.e. Trace of A)

Property : 3 Product of the eigen values of a matrix A is equal to the determinant of A .

Property : 4 If $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are the eigen values of a matrix A , then

(i) $k\lambda_1, k\lambda_2, k\lambda_3, \dots, k\lambda_n$ are the eigen values of the matrix kA , where k is a non-zero scalar.

(ii) $\lambda_1^p, \lambda_2^p, \lambda_3^p, \dots, \lambda_n^p$ are the eigen values of the matrix A^p , where p is a positive integer.

(iii) $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}, \dots, \frac{1}{\lambda_n}$ are the eigen values of the inverse matrix A^{-1} , provided $\lambda_r \neq 0$. (i.e., A is nonsingular)

Property : 5 If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigen values of a matrix A , then $\lambda_1 \pm k, \lambda_2 \pm k, \dots, \lambda_n \pm k$ are the eigenvalues of the matrix $A \pm kI$.

Property : 6 The eigen values of a real symmetric matrix are real.

Property : 7 If zero is a characteristic root of matrix then the matrix is singular.

Properties of Eigen Vectors

Property : 1 The eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.

Illustrative Examples

Example : 1 Find the sum and product of the eigenvalues of $A =$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$$

Solution : Given $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$

Sum of the eigen values

= Sum of the main diagonal elements [by property]

$$= 1 + 3 + 3 = 7$$

Product of the eigen values = determinant of $A = |A|$

$$= 1(9 - 1) - 0 - 0 = 8$$

Example : 2 Find the eigen values A^2 and A^{-1} given matrix $A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

Solution : Given matrix $A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$

Since A is a triangular matrix, the eigen values of A are 2, 3, 5 (the principal diagonal elements of A).

$\therefore$ The eigen values of A^2 are $2^2, 3^2, 5^2$. i.e., 9, 4, 25

$$\text{The eigen values of } A^{-1} \text{ are } \frac{1}{2}, \frac{1}{3}, \frac{1}{5}$$

Example : 3 If 3 and 15 are the two eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$,

find $|A|$ without expanding the determinant.

Solution : Given $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$

Sum of the eigen values

= Sum of the main diagonal elements [by property]

If λ is the third eigen value of A , then

$$3 + 15 + \lambda = 8 + 7 + 3$$

$$\text{i.e., } 18 + \lambda = 18 \Rightarrow \lambda = 0$$

$|A|$ = Product of eigen values of A

$$= (3). (15). (0) = 0$$

Example : 4 Two eigen values of $A = \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{pmatrix}$ are equal and they

are double the third. Find the eigen values of A^2 .

Solution : Given $A = \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{pmatrix}$

Let the third eigen value be λ . Therefore, the other two eigen values are $2\lambda, 2\lambda$

$$\text{i.e., } \lambda + 2\lambda + 2\lambda = 4 + 3 - 2$$

$$\Rightarrow 5\lambda = 5 \Rightarrow \lambda = 1$$

$\therefore$ The eigen values of A are 1, 2, 2

$\therefore$ The eigen values of A^2 are 1, 4, 4

Example : 5 If 3 and 6 are the two eigen values of $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$, write

down all the eigen values of A^{-1} and $3A$.

Solution : Given $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$

Let the third eigen value be λ , then by the property

$$3 + 6 + \lambda = 1 + 5 + 1$$

$$\Rightarrow \lambda = 7 - 9 = -2$$

$\therefore$ The eigen values of A are 3, 6, -2

$\therefore$ The eigen values of A^{-1} are $\frac{1}{3}, \frac{1}{6}, \frac{-1}{2}$ and the eigen values of $3A$ are 9, 18, -6

Example : 6 Find the eigen values of $A - 2I$ and A^{-1} given that two eigen

values of $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ are 2 and 8.

Solution : Given $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

Let the third eigen value be λ , then by the property

$$2 + 8 + \lambda = 6 + 3 + 3$$

$$\lambda = 12 - 10 = 2$$

$\therefore$ The eigen values of A are 2, 2, 8.

$\therefore$ The eigen values of $A - 2I$ are 0, 0, -6 and the eigen values of A^{-1} are

$$\frac{1}{2}, \frac{1}{2}, \frac{1}{8}$$

Example : 7 The product of two eigen values of $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16.

Find the third eigen value.

Solution : Given $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$

Let the third eigen value be $\lambda \Rightarrow$ Product of the eigen values = 16λ

Also, by the property the product of the eigen values = $\det. A = |A| = 32$

$$\therefore 16\lambda = 32 \Rightarrow \lambda = 2$$

$\therefore$ The third eigen value = 2

Example : 8 If -9 is the one of eigen value of $A = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ 4 & -1 & -8 \end{pmatrix}$ then find

the other eigen values.

Solution : Given $A = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ 4 & -1 & -8 \end{pmatrix}$

Let $\lambda_1 = -9$

By the property of eigen values, $\lambda_1 + \lambda_2 + \lambda_3 = 7 - 8 - 8$

$$\Rightarrow -9 + \lambda_2 + \lambda_3 = -9$$

$$\Rightarrow \lambda_2 + \lambda_3 = 0 \Rightarrow \lambda_2 = -\lambda_3$$

$$\lambda_1 \lambda_2 \lambda_3 = |A| \quad [\text{by property}]$$

$$(-9)(-\lambda_3)\lambda_3 = 441$$

$$(\lambda_3)^2 = \frac{441}{9} = 49$$

$$\Rightarrow \lambda_3 = \pm 7$$

$$\therefore \lambda_1 = -9, \lambda_2 = -7, \lambda_3 = 7$$

Problems for Practise

1. Find the sum and product of the eigen values of the matrix $A =$

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

2. Find the eigen value of A^5 , where $A = \begin{pmatrix} 3 & 0 & 0 \\ 5 & 4 & 0 \\ 3 & 6 & 1 \end{pmatrix}$

3. Find the eigen values of $A = \begin{pmatrix} 2 & 3 \\ 0 & 4 \end{pmatrix}$ corresponding to the eigenvector

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Answers

- Sum = 6, Product = 1
- Eigen value of A^5 are $3^5, 4^5, 1^5$
- $\lambda = 2$

Cayley-Hamilton Theorem

Statement : Every square matrix satisfies its own characteristic equation.

This means that if $\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \dots + c_n = 0$ is the characteristic equation of a square matrix A of order n , then $A^n + c_1A^{n-1} + c_2A^{n-2} + \dots + c_nI = 0$

Note : 1 When λ is replaced by A in the characteristic equation, the constant term c_n should be replaced by c_nI to get the result of Cayley-Hamilton theorem, where I is the unit matrix of order n . Also, 0 in the RHS is a null matrix of order n .

Note : 2 Using Cayley-Hamilton theorem we can find A^{-1} and higher positive integral powers of A can be computed.

Note : 3 By Cayley-Hamilton theorem, we can express any higher powers of the matrix as the linear combination of A^2, A & I .

Example : 1 Verify Cayley-Hamilton theorem for the matrix $A =$

$$\begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \text{ and also use it to find } A^{-1}.$$

Solution : Given $A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 1 + 2 + 1 = 4$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 7 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 4 & 2 \end{vmatrix}$$

$$= (2 - 3) + (1 - 7) + (2 - 12) = -4 - 6 - 10 = -20$$

$S_3 = |A|$

$$= \begin{vmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{vmatrix}$$

$$= 1(2 - 6) - 3(4 - 3) + 7(8 - 2)$$

$$= -4 - 3 + 42$$

$$= 35$$

$\therefore$ The characteristic equation is $\lambda^3 - 4\lambda^2 - 20\lambda - 35 = 0$.

The Cayley-Hamilton theorem states that

$$A^3 - 4A^2 - 20A - 35I = 0 \text{ --- (1) which is to be verified.}$$

$$\text{Now, } A^2 = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix}$$

$$A^3 = A.A^2 = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} = \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix}$$

Substituting these values in (1), we get,

$$\text{LHS} = A^3 - 4A^2 - 20A - 35I$$

$$= \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix} - 4 \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} - 20 \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} - 35 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix} - \begin{pmatrix} 80 & 92 & 92 \\ 60 & 88 & 148 \\ 40 & 36 & 56 \end{pmatrix} - \begin{pmatrix} 20 & 60 & 140 \\ 80 & 40 & 60 \\ 20 & 40 & 20 \end{pmatrix} - \begin{pmatrix} 35 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 35 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= 0 = \text{RHS}$$

Thus, Cayley-Hamilton theorem is verified.

To find A^{-1}

Pre-multiplying (1) by A^{-1} , we get

$$A^2 - 4A - 20I - 35A^{-1} = 0$$

$$A^{-1} = \frac{1}{35}(A^2 - 4A - 20I)$$

$$= \frac{1}{35} \left[\begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} - 4 \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} - 20 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right]$$

$$= \frac{1}{35} \left[\begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} - \begin{pmatrix} 4 & 12 & 28 \\ 16 & 8 & 12 \\ 4 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{pmatrix} \right]$$

$$= \frac{1}{35} \begin{pmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{pmatrix}$$

Example : 2 If $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$, Prove that $A^3 - 3A^2 - 9A - 5I = 0$.

Hence find A^4 and A^{-1} .

Solution : Given $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 1 + 1 + 1 = 3$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= (1 - 4) + (1 - 4) + (1 - 4) = -9$$

$S_3 = |A|$

$$= \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix}$$

$$= 1(1 - 4) - 2(2 - 4) + 2(4 - 2) = -3 + 4 + 4 = 5$$

$\therefore$ The characteristic equation is $\lambda^3 - 3\lambda^2 - 9\lambda - 5 = 0$.

By Cayley-Hamilton theorem, we have $A^3 - 3A^2 - 9A - 5I = 0 \dots (1)$

To prove $A^3 - 3A^2 - 9A - 5I = 0$

$$A^2 = A.A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix}$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{pmatrix}$$

$$\therefore \text{RHS} = A^3 - 3A^2 - 9A - 5I$$

$$\begin{aligned} &= \begin{pmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{pmatrix} - 3 \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - 9 \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{pmatrix} - \begin{pmatrix} 27 & 24 & 24 \\ 24 & 27 & 24 \\ 24 & 24 & 27 \end{pmatrix} - \begin{pmatrix} 9 & 18 & 18 \\ 18 & 9 & 18 \\ 18 & 18 & 9 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= 0 = \text{RHS} \end{aligned}$$

$$\therefore A^3 - 3A^2 - 9A - 5I = 0 \text{ --- (1)}$$

To find A^4

Multiplying (1) by A , we get

$$A^4 - 3A^3 - 9A^2 - 5A = 0$$

$$\Rightarrow A^4 = 3A^3 + 9A^2 + 5A$$

$$\begin{aligned} &= 3 \begin{pmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{pmatrix} + 9 \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} + 5 \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 123 & 126 & 126 \\ 126 & 123 & 126 \\ 126 & 126 & 123 \end{pmatrix} + \begin{pmatrix} 81 & 72 & 72 \\ 72 & 81 & 72 \\ 72 & 72 & 81 \end{pmatrix} + \begin{pmatrix} 5 & 10 & 10 \\ 10 & 5 & 10 \\ 10 & 10 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 209 & 208 & 208 \\ 208 & 209 & 208 \\ 208 & 208 & 209 \end{pmatrix} \end{aligned}$$

To find A^{-1}

Multiplying (1) by A^{-1} , we get

$$A^2 - 3A - 9I - 5A^{-1} = 0$$

$$5A^{-1} = A^2 - 3A - 9I$$

$$\begin{aligned} 5A^{-1} &= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - 3 \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 9 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 3 & 6 & 6 \\ 6 & 3 & 6 \\ 6 & 6 & 3 \end{pmatrix} - \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \\ &= \begin{pmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{pmatrix} \\ A^{-1} &= \frac{1}{5} \begin{pmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{pmatrix} \end{aligned}$$

Example : 3 Using Cayley-Hamilton theorem, Find the value of $A^8 - A^7 +$

$$5A^6 - A^5 + A^4 - A^3 + 6A^2 + A - 2I, \text{ if } A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix}$$

Solution :

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 1 + 5 - 5 = 1$$

S_2 = sum of the minors of principal diagonal elements

$$\begin{aligned} &= \begin{vmatrix} 5 & -4 \\ 7 & -5 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ 3 & -5 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} \\ &= (-25 + 28) + (-5 + 6) + (5 - 4) \\ &= 3 + 1 + 1 = 5 \end{aligned}$$

$S_3 = |A|$

$$\begin{aligned} &= \begin{vmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{vmatrix} \\ &= 1(-25 + 28) - 2(-10 + 12) - 2(14 - 15) \\ &= 3 - 4 + 2 = 1 \end{aligned}$$

∴ The characteristic equation is $\lambda^3 - \lambda^2 + 5\lambda - 1 = 0$.

By Cayley-Hamilton theorem, we have $A^3 - A^2 + 5A - I = 0 \dots (1)$

By using the idea of polynomial division,

$$\begin{array}{r}
 A^5 + A \\
 A^3 - A^2 + 5A - I \overline{) A^8 - A^7 + 5A^6 - A^5 + A^4 - A^3 + 6A^2 + A - 2I} \\
 \underline{A^8 - A^7 + 5A^6 - A^5} \\
 A^4 - A^3 + 6A^2 + A - 2I \\
 \underline{A^4 - A^3 + 5A^2 - A} \\
 \text{Remainder} \Rightarrow A^2 + 2A - 2I
 \end{array}$$

$$\therefore A^8 - A^7 + 5A^6 - A^5 + A^4 - A^3 + 6A^2 + A - 2I = A^2 + 2A - 2I \dots (1)$$

$$\text{Now, } A^2 = A \cdot A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix} = \begin{pmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{pmatrix}$$

$$\begin{aligned}
 A^2 + 2A - 2I &= \begin{pmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{pmatrix} + 2 \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} -1 & -2 & 0 \\ 0 & 1 & -4 \\ 2 & 6 & -9 \end{pmatrix} + \begin{pmatrix} 2 & 4 & -4 \\ 4 & 10 & -8 \\ 6 & 14 & -10 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} -1 & 2 & -4 \\ 4 & 9 & -12 \\ 8 & 20 & -21 \end{pmatrix}
 \end{aligned}$$

∴ From (1) $\Rightarrow$

$$A^8 - A^7 + 5A^6 - A^5 + A^4 - A^3 + 6A^2 + A - 2I = \begin{pmatrix} -1 & 2 & -4 \\ 4 & 9 & -12 \\ 8 & 20 & -21 \end{pmatrix}$$

Problems for Practise

1. Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{pmatrix}$. Hence express

A^7 in terms of A^2, A & I

2. Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 3 & -1 \\ -1 & 5 \end{pmatrix}$

3. Use Cayley-Hamilton theorem to find the inverse of $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$

4. Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$ and

also use it to find A^{-1}

5. Verify that the matrix $A = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ satisfies its characteristic

equation and hence find A^4

6. Use Cayley-Hamilton theorem to find the inverse of $A = \begin{pmatrix} -1 & 0 & 3 \\ 8 & 1 & -7 \\ -3 & 0 & 8 \end{pmatrix}$

Answers

1. To verify $A^3 - A^2 + 5A - I = 0$, $A^7 = 67A^2 - 73A + 13I$

2. To verify $A^2 - 8A + 14I = 0$

3. Characteristic Equation is $\lambda^2 - 4\lambda - 5 = 0$, $A^{-1} = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 2 & -1 \end{pmatrix}$

4. To Verify $A^3 - 4A^2 - 20A - 35I = 0$, $A^{-1} = \frac{1}{35} \begin{pmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{pmatrix}$

5. To Verify $A^3 - 6A^2 + 8A - 3I = 0$, $A^4 = \begin{pmatrix} 124 & -123 & 162 \\ -95 & 96 & -123 \\ 95 & -95 & 124 \end{pmatrix}$

6. Characteristic Equation is $A^3 - 8A^2 + 8A - I = 0$

Orthogonal Matrix

A square matrix A is called orthogonal if

$$AA^T = A^T A = I$$

Also, $AA^{-1} = A^{-1}A = I$

Hence, for an orthogonal matrix, $A^{-1} = A^T$

Pair wise Orthogonal

If $X_1X_2^T = 0$, $X_2X_3^T = 0$ and $X_3X_1^T = 0$, then the eigen vectors X_1, X_2, X_3 are said to be pair wise orthogonal.

Similarity Transformation

If A and B are two square matrices of order n then B is said to be similar to A , if there exists a non singular matrix P such that $B = P^{-1}AP$

Diagonalisation of a Matrix

Definition : The process of finding a matrix M such that $M^{-1}AM = D$, where D is a diagonal matrix is called diagonalisation of the matrix A .

Since, $M^{-1}AM = D$, A and D are similar matrices and therefore A and D have the same eigen values. It is clear that the diagonal matrix D has eigen values of A as its diagonal elements.

The transformation $M^{-1}AM = D$ is called a similarity transformation.

Calculation of powers of a matrix

If a square matrix A has linearly independent eigen vectors, then by the above definition,

$$D = M^{-1}AM$$

$$\therefore A = MDM^{-1}$$

$$A^2 = (MDM^{-1})(MDM^{-1})$$

$$= MD(M^{-1}M)DM^{-1}$$

$$= MD^2M^{-1}$$

Similarly, $A^3 = MD^3M^{-1}$

Proceeding, we get $A^k = MD^kM^{-1}$

Illustrative Examples

Example:1 Diagonalise the matrix $A = \begin{pmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix}$ by similarity

transformation and hence find A^4 .

Solution : Given $A = \begin{pmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix}$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 2 + 1 - 3 = 0$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 1 & 2 \\ 1 & -3 \end{vmatrix} + \begin{vmatrix} 2 & -7 \\ 0 & -3 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= (-3 - 2) + (-6 - 0) + (2 - 4)$$

$$= -5 - 6 - 2 = -13$$

$$S_3 = |A|$$

$$= \begin{vmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{vmatrix}$$

$$= 2(-3-2) - 2(-6-0) - 7(2-0)$$

$$= -10 + 12 - 14$$

$$= -12$$

∴ The characteristic equation is $\lambda^3 - 0\lambda^2 - 13\lambda + 12 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -13 & 12 \\ & & 0 & 1 & 1 \\ \hline & 1 & 1 & -12 & 0 \end{array}$$

i.e., $\lambda = 1, \lambda^2 + \lambda - 12 = 0$

i.e., $\lambda = 1, (\lambda - 3)(\lambda + 4) = 0$

∴ The eigenvalues are $\lambda = 1, 3, -4$

To find the eigenvector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} 2-\lambda & 2 & -7 \\ 2 & 1-\lambda & 2 \\ 0 & 1 & -3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (2-\lambda)x_1 + 2x_2 - 7x_3 &= 0 \\ 2x_1 + (1-\lambda)x_2 + 2x_3 &= 0 \\ 0x_1 + 1x_2 + (-3-\lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = 1$

(1) $\Rightarrow$ becomes, $1x_1 + 2x_2 - 7x_3 = 0$

$2x_1 + 0x_2 + 2x_3 = 0$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 2 & -7 & 1 \\ \swarrow & \searrow & \swarrow \\ 0 & 2 & 2 \end{array} \quad \begin{array}{ccc} 1 & 2 & 0 \\ \swarrow & \searrow & \swarrow \\ 2 & 2 & 0 \end{array}$$

$$\frac{x_1}{4-0} = \frac{x_2}{-14-2} = \frac{x_3}{0-4}$$

$$\frac{x_1}{4} = \frac{x_2}{-16} = \frac{x_3}{-4}$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{-4} = \frac{x_3}{-1}$$

$$X_1 = \begin{pmatrix} 1 \\ -4 \\ -1 \end{pmatrix}$$

Case (ii) : $\lambda = 3$

(1) $\Rightarrow$ becomes, $-1x_1 + 2x_2 - 7x_3 = 0$

$2x_1 - 2x_2 + 2x_3 = 0$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 2 & -7 & -1 \\ \swarrow & \searrow & \swarrow \\ -2 & 2 & 2 \end{array} \quad \begin{array}{ccc} -1 & 2 & 2 \\ \swarrow & \searrow & \swarrow \\ 2 & -2 & -2 \end{array}$$

$$\frac{x_1}{4-14} = \frac{x_2}{-14+2} = \frac{x_3}{2-4}$$

$$\frac{x_1}{-10} = \frac{x_2}{-12} = \frac{x_3}{-2} \Rightarrow \frac{x_1}{5} = \frac{x_2}{6} = \frac{x_3}{1}$$

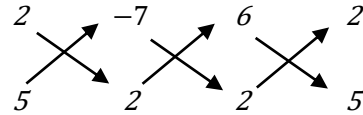
$$X_2 = \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix}$$

Case (iii) : $\lambda = -4$

(1) $\Rightarrow$ becomes, $6x_1 + 2x_2 - 7x_3 = 0$

$2x_1 + 5x_2 + 2x_3 = 0$

Solving, by the rule of cross-multiplication, we have



$$\frac{x_1}{4+35} = \frac{x_2}{-14-12} = \frac{x_3}{30-4}$$

$$\frac{x_1}{39} = \frac{x_2}{-26} = \frac{x_3}{26} \Rightarrow \frac{x_1}{3} = \frac{x_2}{-2} = \frac{x_3}{2}$$

$$X_3 = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$$

Hence the modal matrix is $M = \begin{pmatrix} 1 & 5 & 3 \\ -4 & 6 & -2 \\ -1 & 1 & 2 \end{pmatrix}$.

We know that, $M^{-1} = \frac{1}{|M|} \text{adj } M$, where $\text{adj } M = (\text{cofactor of } M)^T$

$$|M| = 1(12+2) - 5(-8-2) + 3(-4+6)$$

$$= 14 + 50 + 6 = 70$$

To find cofactor of M

Let Cofactor of $M = \begin{pmatrix} a_{11} & -a_{12} & a_{13} \\ -a_{21} & a_{22} & -a_{23} \\ a_{31} & -a_{32} & a_{33} \end{pmatrix} \dots (1)$

$$a_{11} = \begin{vmatrix} 6 & -2 \\ 1 & 2 \end{vmatrix} = 12 + 2 = 14$$

$$a_{12} = \begin{vmatrix} -4 & -2 \\ -1 & 2 \end{vmatrix} = -8 - 2 = -10$$

$$a_{13} = \begin{vmatrix} -4 & 6 \\ -1 & 1 \end{vmatrix} = -4 + 6 = 2$$

$$a_{21} = \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = 10 - 3 = 7$$

$$a_{22} = \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} = 2 + 3 = 5$$

$$a_{23} = \begin{vmatrix} 1 & 5 \\ -1 & 1 \end{vmatrix} = 1 + 5 = 6$$

$$a_{31} = \begin{vmatrix} 5 & 3 \\ 6 & -2 \end{vmatrix} = -10 - 18 = -28$$

$$a_{32} = \begin{vmatrix} 1 & 3 \\ -4 & -2 \end{vmatrix} = -2 + 12 = 10$$

$$a_{33} = \begin{vmatrix} 1 & 5 \\ -4 & 6 \end{vmatrix} = 6 + 20 = 26$$

$$(1) \Rightarrow \text{Cofactor of } M = \begin{pmatrix} 14 & 10 & 2 \\ -7 & 5 & -6 \\ -28 & -10 & 26 \end{pmatrix}$$

$$\text{Now, } \text{adj } M = (\text{cofactor of } M)^T = \begin{pmatrix} 14 & 10 & 2 \\ -7 & 5 & -6 \\ -28 & -10 & 26 \end{pmatrix}^T$$

$$\text{adj } M = \begin{pmatrix} 14 & -7 & -28 \\ 10 & 5 & -10 \\ 2 & -6 & 26 \end{pmatrix}$$

$$\text{Therefore, } M^{-1} = \frac{1}{70} \begin{pmatrix} 14 & -7 & -28 \\ 10 & 5 & -10 \\ 2 & -6 & 26 \end{pmatrix}$$

$$\text{Now, } D = M^{-1}AM$$

$$AM = \begin{pmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 5 & 3 \\ -4 & 6 & -2 \\ -1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 15 & -12 \\ -4 & 18 & 8 \\ -1 & 3 & -8 \end{pmatrix}$$

$$M^{-1}AM = \frac{1}{70} \begin{pmatrix} 14 & -7 & -28 \\ 10 & 5 & -10 \\ 2 & -6 & 26 \end{pmatrix} \begin{pmatrix} 1 & 15 & -12 \\ -4 & 18 & 8 \\ -1 & 3 & -8 \end{pmatrix}$$

$$= \frac{1}{70} \begin{pmatrix} 70 & 0 & 0 \\ 0 & 210 & 0 \\ 0 & 0 & -280 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -4 \end{pmatrix}$$

A^4 is given by

$$A^4 = MD^4M^{-1}$$

$$D^4M^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 256 \end{pmatrix} \times \frac{1}{70} \begin{pmatrix} 14 & -7 & -28 \\ 10 & 5 & -10 \\ 2 & -6 & 26 \end{pmatrix}$$

$$= \frac{1}{70} \begin{pmatrix} 14 & -7 & -28 \\ 810 & 405 & -810 \\ 512 & -1536 & 6656 \end{pmatrix}$$

$$MD^4M^{-1} = \begin{pmatrix} 1 & 5 & 3 \\ -4 & 6 & -2 \\ -1 & 1 & 2 \end{pmatrix} \times \frac{1}{70} \begin{pmatrix} 14 & -7 & -28 \\ 810 & 405 & -810 \\ 512 & -1536 & 6656 \end{pmatrix}$$

$$= \frac{1}{70} \begin{pmatrix} 5600 & -2590 & 15890 \\ 3780 & 5530 & -18060 \\ 1820 & -2660 & 12530 \end{pmatrix}$$

$$A^4 = \begin{pmatrix} 80 & -37 & 227 \\ 54 & 79 & -258 \\ 26 & -38 & 179 \end{pmatrix}$$

Diagonalisation of a Real Symmetric Matrix by Orthogonal Reduction

Definition : If $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ is a column vector, then the normalized vector of

$$X = \begin{pmatrix} \frac{x_1}{\sqrt{x_1^2 + x_2^2 + x_3^2}} \\ \frac{x_2}{\sqrt{x_1^2 + x_2^2 + x_3^2}} \\ \frac{x_3}{\sqrt{x_1^2 + x_2^2 + x_3^2}} \end{pmatrix}$$

Let A be a real symmetric matrix. Since A is a real symmetric matrix, the eigen vectors of A are not only linearly independent but also pairwise orthogonal.

Let N be the matrix whose columns are the normalized eigen vectors of A . Then N is an orthogonal matrix and by property $N^{-1} = N^T$.

The matrix N is called the normalized modal matrix. Then the transformation $D = N^{-1}AN = N^TAN$ is known as orthogonal transformation or orthogonal reduction. Thus, the square matrix A can be diagonalised by an orthogonal reduction $N^TAN = D$.

ILLUSTRATIVE EXAMPLES

Example : 1 Reduce the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$ to a diagonal form by orthogonal reduction.

Solution :

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 8 + 7 + 3 = 18$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}$$

$$= (21 - 16) + (24 - 4) + (56 - 36)$$

$$= 5 + 20 + 20 = 45$$

$S_3 = |A|$

$$= \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix}$$

$$= 8(21 - 16) + 6(-18 + 8) + 2(24 - 14) = 40 - 60 + 20 = 0$$

$\therefore$ The characteristic equation is $\lambda^3 - 18\lambda^2 + 45\lambda = 0$

i. e., $\lambda(\lambda^2 - 18\lambda + 45) = 0$

$$\lambda(\lambda - 3)(\lambda - 15) = 0$$

i. e., $\lambda = 0, 3, 15$

$\therefore$ The eigenvalues are $\lambda = 0, 3, 15$

To find the eigenvector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (8-\lambda)x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + (7-\lambda)x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + (3-\lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = 0$

(1) $\Rightarrow$ becomes, $8x_1 - 6x_2 + 2x_3 = 0$
 $-6x_1 + 7x_2 - 4x_3 = 0$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -6 & 2 & 8 \\ 7 & -4 & -6 \end{array}$$

$$\frac{x_1}{24-14} = \frac{x_2}{-12+32} = \frac{x_3}{56-36}$$

$$\frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{2}$$

$$X_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

Case (ii) : $\lambda = 3$

(1) $\Rightarrow$ becomes, $5x_1 - 6x_2 + 2x_3 = 0$
 $-6x_1 + 4x_2 - 4x_3 = 0$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -6 & 2 & 5 \\ 4 & -4 & -6 \end{array}$$

$$\frac{x_1}{24-8} = \frac{x_2}{-12+20} = \frac{x_3}{20-36}$$

$$\frac{x_1}{16} = \frac{x_2}{8} = \frac{x_3}{-16} \Rightarrow \frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{-2}$$

$$X_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Case (iii) : $\lambda = 15$

(1) $\Rightarrow$ becomes, $-7x_1 - 6x_2 + 2x_3 = 0$
 $-6x_1 - 8x_2 - 4x_3 = 0$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -6 & 2 & -7 \\ -8 & -4 & -6 \end{array}$$

$$\frac{x_1}{24+16} = \frac{x_2}{-12-28} = \frac{x_3}{56-36}$$

$$\frac{x_1}{40} = \frac{x_2}{-40} = \frac{x_3}{20} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{1}$$

$$X_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

The eigen vectors are $X_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, $X_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$, $X_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$

Clearly, X_1, X_2, X_3 are pairwise orthogonal.

Consider the matrix N whose columns are normalized eigen vectors.

$$\therefore N = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

$$N^T A N = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{3}{3} & \frac{3}{3} & -\frac{3}{3} \end{pmatrix} \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{3}{3} & \frac{3}{3} & -\frac{3}{3} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix} = D$$

Hence A is reduced to a diagonal matrix by means of orthogonal reduction.

Example : 2 Find an orthogonal matrix which reduces the matrix $A =$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \text{ to a diagonal matrix.}$$

Solution :

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

$S_1 =$ sum of the principal diagonal elements

$$= 1 + 5 + 1 = 7$$

$S_2 =$ sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 3 & 5 \end{vmatrix}$$

$$= (5 - 1) + (1 - 9) + (5 - 1)$$

$$= 4 - 8 + 4 = 0$$

$S_3 = |A|$

$$= \begin{vmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{vmatrix}$$

$$= 1(5 - 1) - 1(1 - 3) + 3(1 - 15) = 4 + 2 - 42 = -36$$

$\therefore$ The characteristic equation is $\lambda^3 - 7\lambda^2 + 36 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} -2 & 1 & -7 & 0 & 36 \\ & & -2 & 18 & -36 \\ \hline & 1 & -9 & 18 & 0 \end{array}$$

$$\text{i.e., } \lambda = -2, \lambda^2 - 9\lambda + 18 = 0$$

$$\text{i.e., } \lambda = -2, \quad (\lambda - 3)(\lambda - 6) = 0$$

$\therefore$ The eigenvalues are $\lambda = -2, 3, 6$

To find the eigenvector

$$\text{Let } (A - \lambda I)X = 0, \text{ where } X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 - \lambda & 1 & 3 \\ 1 & 5 - \lambda & 1 \\ 3 & 1 & 1 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (1 - \lambda)x_1 + 1x_2 + 3x_3 &= 0 \\ 1x_1 + (5 - \lambda)x_2 + 1x_3 &= 0 \\ 3x_1 + 1x_2 + (1 - \lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = -2$

$$(1) \Rightarrow \text{becomes, } 3x_1 + 1x_2 + 3x_3 = 0$$

$$1x_1 + 7x_2 + 1x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 3 & 3 \\ \swarrow & \searrow & \swarrow \\ 7 & 1 & 1 \end{array}$$

$$\frac{x_1}{1 - 21} = \frac{x_2}{3 - 3} = \frac{x_3}{21 - 1}$$

$$\frac{x_1}{-20} = \frac{x_2}{0} = \frac{x_3}{20} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{0} = \frac{x_3}{1}$$

$$X_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Case (ii) : $\lambda = 3$

$$(1) \Rightarrow \text{becomes, } -2x_1 + 1x_2 + 3x_3 = 0$$

$$1x_1 + 2x_2 + 1x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 1 & 3 & -2 \\ 2 & 1 & 1 \end{array} \quad \begin{array}{ccc} 3 & -2 & 1 \\ 1 & 1 & 2 \end{array} \quad \begin{array}{ccc} -2 & 1 & 1 \\ 1 & 2 & 1 \end{array}$$

$$\frac{x_1}{1-6} = \frac{x_2}{3+2} = \frac{x_3}{-4-1}$$

$$\frac{x_1}{-5} = \frac{x_2}{5} = \frac{x_3}{-5} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{1} = \frac{x_3}{-1}$$

$$X_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

Case (iii) : The third eigen vector is given by pairwise orthogonal conditions

$$X_1^T X_3 = 0 ; X_2^T X_3 = 0$$

$$\text{Let } X_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\therefore X_1^T X_3 = 0 \Rightarrow (-1 \ 0 \ 1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow -x_1 + 0x_2 + x_3 = 0$$

$$\& X_2^T X_3 = 0 \Rightarrow (-1 \ 1 \ -1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow -x_1 + x_2 - x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} 0 & 1 & -1 \\ 1 & -1 & -1 \end{array} \quad \begin{array}{ccc} 1 & -1 & 0 \\ -1 & -1 & 1 \end{array} \quad \begin{array}{ccc} -1 & 0 & 0 \\ -1 & 1 & -1 \end{array}$$

$$\frac{x_1}{0-1} = \frac{x_2}{-1-1} = \frac{x_3}{-1-0}$$

$$\frac{x_1}{-1} = \frac{x_2}{-2} = \frac{x_3}{-1} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{1}$$

$$X_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Consider the matrix N whose columns are normalized eigen vectors

$$\therefore N = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{pmatrix}$$

Now,

$$\begin{aligned} N^T A N &= \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{pmatrix} \\ &= \begin{pmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} \\ &= D \end{aligned}$$

Problems for Practise

1. Diagonalise the matrix $A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{pmatrix}$ by means of orthogonal transformation.

2. Diagonalise the matrix $A = \begin{pmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{pmatrix}$ by means of orthogonal transformation.

3. Diagonalise the matrix $A = \begin{pmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{pmatrix}$ by means of orthogonal transformation.

Answers

$$1. N = \begin{pmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \end{pmatrix}, D = (-1, 1, 4)$$

$$2. N = \begin{pmatrix} \frac{1}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} \\ \frac{-5}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} \\ \frac{4}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} \end{pmatrix}, D = (0, 3, 14)$$

$$3. N = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}, D = (-2, 6, 6)$$

Quadratic forms

Definition : A homogeneous polynomial of the second degree in any number of variables is called a **quadratic form**.

For example, Quadratic forms of two variables:

$$x^2 + 4xy - 2y^2, x_1^2 - 3x_1x_2 + 2x_2^2$$

Quadratic forms of three variables:

$$x^2 + 2y^2 - 3z^2 + 4xy - 6yz + 8zx, \\ 3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_1x_2 + 2x_1x_3 - 2x_2x_3$$

General form

The general form of a quadratic form in n variables $x_1, x_2, \dots, x_n$ is denoted by $Q = \sum_{j=1}^n \sum_{i=1}^n a_{ij} \cdot x_i \cdot x_j$, where $a_{ij} = a_{ji}$.

The matrix of the coefficients $A = [a_{ij}]$ is a symmetric matrix, called the matrix of the quadratic form. The general form $Q = \sum_{j=1}^n \sum_{i=1}^n a_{ij} \cdot x_i \cdot x_j$ where $a_{ij} = a_{ji}$ can always be written as $X^T A X$, where

$$A = (a_{ij}), X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \text{ and } X^T = (x_1 \quad x_2 \quad \dots \quad x_n)$$

For example,

$$Q = x_1^2 - 4x_1x_2 + 3x_2^2, \text{ can be written}$$

$$Q = (x_1 x_2) \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ = X^T A X$$

$$\text{Where } X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ and } A = \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix}$$

Note: 1 To find the symmetric matrix A of a quadratic form,

- the coefficients a_{ij} are the coefficients of x_i^2 in the quadratic form
- half of coefficient $x_i x_j$ is placed in each of the positions a_{ij} and a_{ji} of A.

Note: 2 (short cut method)

For three variables x, y, z first write

$$\begin{pmatrix} x & y & z \\ x & y & z \\ x & y & z \end{pmatrix} \text{ and then}$$

$$\begin{pmatrix} x & y & z \\ x & y & z \\ x & y & z \end{pmatrix} \Rightarrow \text{multiply first row by } x$$

$$\Rightarrow \text{multiply second row by } y$$

$$\Rightarrow \text{multiply third row by } z$$

That is,
$$\begin{pmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{pmatrix}$$

Similarly we can write for x_1, x_2, x_3

Examples

(i) The matrix of the quadratic form $x^2 - y^2 - z^2 + 4xy + 6xz - 2yz$ is

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & -1 \\ 3 & -1 & -1 \end{pmatrix}$$

(ii) The matrix of the quadratic form $x_1^2 + x_2^2 + 3x_3^2 + 3x_1x_2 + 3x_1x_3 + 3x_2x_3$ is

$$\begin{pmatrix} 1 & 3/2 & 3/2 \\ 3/2 & 1 & 3/2 \\ 3/2 & 3/2 & 3 \end{pmatrix}$$

(iii) Similarly, we reverse the process that is the quadratic form whose

matrix $A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ is $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz + 2xz$

Canonical form of a quadratic form

A real quadratic form $Q = X^TAX$ be reduced as sum or difference of the squares of the new variables, then the resulting quadratic expression is called the Canonical form of the given quadratic form.

In the linear transformation, $X = PY$, the matrix P is chosen such that $B = P^TAP = \text{diag}(d_1, d_2, \dots, d_n)$, a diagonal matrix, then the quadratic form $d_1y_1^2 + d_2y_2^2 + \dots + d_ny_n^2$, which is called the **canonical form** or **sum of squares form of Q** .

Nature of the quadratic form

Let $Q = X^TAX$ be a quadratic form in n variables $x_1, x_2, \dots, x_n$.

Rank of the quadratic form (r)

If the canonical form of Q consists only ' r ' square terms. Then, the rank of A is r .

Index of the quadratic form (s)

The number of positive square terms in the canonical form is called the index of the quadratic form. Therefore, the number of positive square terms is index (s).

Signature of the quadratic form

The difference of number of positive and negative square terms is called the signature of the quadratic form.

Nature of the quadratic form

The quadratic form $Q = X^TAX$ in n variables is said to be

(i) **Positive definite** if $r = n$ and $s = n$

(or) all the eigen values of A are positive numbers.

(ii) **Negative definite** if $r = n$ and $s = 0$

(or) all the eigen values of A are negative numbers.

(iii) **Positive semi definite** if $r < n$ and $s = r$

(or) all the eigen values of A are positive numbers and atleast one eigen value is zero.

(iv) **Negative semi definite** if $r < n$ and $s = 0$

(or) all the eigen values of A are negative numbers and at least one eigen value is zero.

(v) **Indefinite** in all other cases

(or) if A has both positive and negative eigen values.

Test for nature of a quadratic form through principal minors

Let $A = [a_{ij}]$ be the matrix of the quadratic form in n variables. Then A is a square symmetric matrix of order n .

$$\text{Let } D_1 = |a_{11}| = a_{11}$$

$$D_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$D_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\dots, D_n = |A|$$

Where $D_1, D_2, D_3, \dots, D_n$ are the principal minors of A .

- The quadratic form is positive definite if $D_1, D_2, D_3, \dots, D_n$ are all positive. i.e. $D_n > 0$, for all n .
- The quadratic form is negative definite if $D_1, D_3, D_5, \dots$ are all negative and $D_2, D_4, D_6, \dots$ are all positive. i.e. $(-1)^n D_n > 0$, for all n .

- The quadratic form is positive semi definite if $D_n \geq 0$ and atleast one $D_i = 0$.
- The quadratic form is negative semi definite if $(-1)^n D_n \geq 0$ and atleast one $D_i = 0$.
- The quadratic form is indefinite in all other cases.

Reduction of a quadratic form by means of orthogonal reduction

Consider a quadratic form $Q = X^T A X$ in n variables $x_1, x_2, \dots, x_n$.

- ❖ Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigen values of A and $E_1, E_2, \dots, E_n$ be the normalized eigen vectors of A .
- ❖ Consider the matrix N whose columns are normalized eigen vectors of A , called normalized modal matrix.
- ❖ Clearly N is an orthogonal matrix.
- ❖ Then the transformation $X = NY$ where Y is a column vector of new variables $Y_1, Y_2, \dots, Y_n$ reduces the given quadratic form Q into $Y^T D Y$ where D is a diagonal matrix with the eigen values of A as diagonal elements.
- ❖ The resulting Q.F $Y^T D Y$ takes the form $\lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$, which is the required canonical form.
- ❖ This is applied only when the eigen vectors of A are pairwise orthogonal.

ILLUSTRATIVE EXAMPLES

Example : 1 Write the matrix of the quadratic form $2x_1^2 - 2x_2^2 + 4x_3^2 + 2x_1x_2 - 6x_1x_3 + 6x_2x_3$

Solution : Matrix of the quadratic form

$$A = \begin{pmatrix} \text{coeff. of } x_1^2 & \frac{1}{2} \text{coeff. of } x_1x_2 & \frac{1}{2} \text{coeff. of } x_1x_3 \\ \frac{1}{2} \text{coeff. of } x_2x_1 & \text{coeff. of } x_2^2 & \frac{1}{2} \text{coeff. of } x_2x_3 \\ \frac{1}{2} \text{coeff. of } x_3x_1 & \frac{1}{2} \text{coeff. of } x_3x_2 & \text{coeff. of } x_3^2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 1 & -3 \\ 1 & -2 & 3 \\ -3 & 3 & 4 \end{pmatrix}$$

Example : 2 Write the matrix of the quadratic form $10x_1^2 + 2x_2^2 + 5x_3^2 + 6x_2x_3 - 10x_3x_1 - 4x_1x_2$

Solution : Matrix of the quadratic form $A =$

$$\begin{pmatrix} \text{coeff. of } x_1^2 & \frac{1}{2} \text{coeff. of } x_1x_2 & \frac{1}{2} \text{coeff. of } x_1x_3 \\ \frac{1}{2} \text{coeff. of } x_2x_1 & \text{coeff. of } x_2^2 & \frac{1}{2} \text{coeff. of } x_2x_3 \\ \frac{1}{2} \text{coeff. of } x_3x_1 & \frac{1}{2} \text{coeff. of } x_3x_2 & \text{coeff. of } x_3^2 \end{pmatrix} = \begin{pmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{pmatrix}$$

Example : 3 Write the quadratic form corresponding to the following

symmetric matrix $\begin{pmatrix} 0 & -1 & 2 \\ -1 & 1 & 4 \\ 2 & 4 & 3 \end{pmatrix}$

Solution : Quadratic form corresponding to the symmetric matrix A is

$$X'AX = (x_1 \ x_2 \ x_3) \begin{pmatrix} 0 & -1 & 2 \\ -1 & 1 & 4 \\ 2 & 4 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= (0x_1 - x_2 + 2x_3 \quad -x_1 + x_2 + 4x_3 \quad 2x_1 + 4x_2 + 3x_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= (0x_1 - x_2 + 2x_3)x_1 + (-x_1 + x_2 + 4x_3)x_2 + (2x_1 + 4x_2 + 3x_3)x_3$$

$$= 0x_1^2 - x_2x_1 + 2x_3x_1 - x_1x_2 + x_2^2 + 4x_3x_2 + 2x_1x_3 + 4x_2x_3 + 3x_3^2$$

$$= 0x_1^2 + x_2^2 + 3x_3^2 - x_2x_1 - 2x_1x_2 + 4x_1x_3 + 8x_2x_3$$

Problems for Practise

- Find the matrix of the quadratic form $2x_1^2 - 3x_1x_2 + 4x_2^2$
- Write the quadratic form corresponding to the matrix $\begin{pmatrix} 1 & 2 & 5 \\ 2 & 4 & 6 \\ 5 & 6 & 3 \end{pmatrix}$
- Write the quadratic form corresponding to the matrix $\begin{pmatrix} 3 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 6 \\ 0 & 6 & -7 \end{pmatrix}$

Answers

- $\begin{pmatrix} 2 & \frac{-3}{2} \\ \frac{-3}{2} & 4 \end{pmatrix}$
- $x_1^2 + 4x_2^2 + 3x_3^2 + 4x_1x_2 + 12x_2x_3 + 10x_1x_3$
- $3x_1^2 - 7x_3^2 + x_1x_2 + 12x_2x_3$

ILLUSTRATIVE EXAMPLES

Example : 1 Determine the nature of the Q.F $x^2 + 2y^2 + 3z^2 + 2xy + 2yz - 2xz$ without reducing it to canonical form

Solution : The matrix A of the Q.F is $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{pmatrix}$

$$D_1 = |1| = 1; \quad D_2 = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1; \quad D_3 = \begin{vmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{vmatrix} = -2$$

$$\therefore D_1 = +ve; \quad D_2 = +ve; \quad D_3 = -ve$$

$\therefore$ The given Q.F is indefinite.

Example : 2 Determine the nature of the Q.F $5x^2 + 26y^2 + 10z^2 + 4yz + 6xy + 14zx$ using principal minors

Solution : The matrix A of the Q.F is $A = \begin{pmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{pmatrix}$

$$D_1 = |5| = 5; D_2 = \begin{vmatrix} 5 & 3 \\ 3 & 26 \end{vmatrix} = 121; D_3 = \begin{vmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{vmatrix} = 0$$

$$\therefore D_1 = +ve; D_2 = +ve; D_3 = 0$$

$\therefore$ The given Q.F is positive semi-definite.

Example : 3 Determine the nature of the Q.F $x_1^2 + 3x_2^2 + 6x_3^2 + 2x_2x_3 + 4x_3x_1 + 2x_1x_2$ without reducing it to canonical form

Solution : The matrix A of the Q.F is $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 6 \end{pmatrix}$

$$D_1 = |1| = 1; D_2 = \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 2; D_3 = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 6 \end{vmatrix} = 3$$

$$\therefore D_1 = +ve; D_2 = +ve; D_3 = +ve$$

$\therefore$ The given Q.F is positive definite.

Problems for Practise

Determine the nature of the following Quadratic forms:

1. $2x_1^2 + x_2^2 - 3x_3^2 + 12x_1x_2 - 8x_2x_3 - 4x_3x_1$
2. $6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_1x_3 + 4x_1x_2$
3. $x_1^2 - 2x_1x_2 + x_2^2 + x_3^2$

Answers

1. Indefinite
2. +ve definite
3. +ve semi-definite

ILLUSTRATIVE EXAMPLES

Example :1 By an orthogonal transformation, reduce the Q.F $3x^2 - 2y^2 - z^2 - 4xy + 12yz + 8xz$ to a canonical form. Find also its nature.

Solution : The given Q.F can be rewritten in matrix form as $A =$

$$\begin{pmatrix} 3 & -2 & 4 \\ -2 & -2 & 6 \\ 4 & 6 & -1 \end{pmatrix}$$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

$S_1 =$ sum of the principal diagonal elements

$$= 3 - 2 - 1 = 0$$

$S_2 =$ sum of the minors of principal diagonal elements

$$= \begin{vmatrix} -2 & 6 \\ 6 & -1 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 4 & -1 \end{vmatrix} + \begin{vmatrix} 3 & -2 \\ -2 & -2 \end{vmatrix} \\ = (2 - 36) + (-3 - 16) + (-6 - 4) = -34 - 19 - 10 = -63$$

$S_3 = |A|$

$$= \begin{vmatrix} 3 & -2 & 4 \\ -2 & -2 & 6 \\ 4 & 6 & -1 \end{vmatrix} \\ = 3(2 - 36) + 2(2 - 24) + 4(-12 + 8) \\ = -102 - 44 - 16 = -162$$

$\therefore$ The characteristic equation is $\lambda^3 - 63\lambda + 162 = 0$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} 3 & 1 & 0 & -63 & 162 \\ & & 3 & 9 & -162 \\ \hline & 1 & 3 & -54 & 0 \end{array}$$

$$\text{i.e., } \lambda = 3, \lambda^2 + 3\lambda - 54 = 0$$

$$\text{i.e., } \lambda = 3, (\lambda - 6)(\lambda + 9) = 0$$

∴ The eigenvalues are $\lambda = 3, 6, -9$

To find the eigenvector

Let $(A - \lambda I)X = 0$, where $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$\begin{pmatrix} 3-\lambda & -2 & 4 \\ -2 & -2-\lambda & 6 \\ 4 & 6 & -1-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (3-\lambda)x_1 - 2x_2 + 4x_3 &= 0 \\ -2x_1 + (-2-\lambda)x_2 + 6x_3 &= 0 \\ 4x_1 + 6x_2 + (-1-\lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = 3$

$$(1) \Rightarrow \text{becomes, } \begin{aligned} 0x_1 - 2x_2 + 4x_3 &= 0 \\ -2x_1 - 5x_2 + 6x_3 &= 0 \end{aligned}$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -2 & 4 & 0 \\ \swarrow & \searrow & \swarrow \\ -5 & 6 & -2 \end{array} \quad \begin{array}{ccc} 4 & 0 & -2 \\ \swarrow & \searrow & \swarrow \\ 6 & -2 & -5 \end{array}$$

$$\frac{x_1}{-12+20} = \frac{x_2}{-8-0} = \frac{x_3}{0-4}$$

$$\frac{x_1}{8} = \frac{x_2}{-8} = \frac{x_3}{-4} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{-1}$$

$$X_1 = \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}$$

Case (ii) : $\lambda = 6$

$$(1) \Rightarrow \text{becomes, } \begin{aligned} -3x_1 - 2x_2 + 4x_3 &= 0 \\ -2x_1 - 8x_2 + 6x_3 &= 0 \end{aligned}$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -2 & 4 & -3 \\ \swarrow & \searrow & \swarrow \\ -8 & 6 & -2 \end{array} \quad \begin{array}{ccc} 4 & -3 & -2 \\ \swarrow & \searrow & \swarrow \\ 6 & -2 & -8 \end{array}$$

$$\frac{x_1}{-12+32} = \frac{x_2}{-8+18} = \frac{x_3}{24-4}$$

$$\frac{x_1}{20} = \frac{x_2}{10} = \frac{x_3}{20} \Rightarrow \frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{2}$$

$$X_2 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

Case (iii) : $\lambda = -9$

$$(1) \Rightarrow \text{becomes, } \begin{aligned} 12x_1 - 2x_2 + 4x_3 &= 0 \\ -2x_1 + 7x_2 + 6x_3 &= 0 \end{aligned}$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -2 & 4 & 12 \\ \swarrow & \searrow & \swarrow \\ 7 & 6 & -2 \end{array} \quad \begin{array}{ccc} 4 & 12 & -2 \\ \swarrow & \searrow & \swarrow \\ 6 & -2 & 7 \end{array}$$

$$\frac{x_1}{-12-28} = \frac{x_2}{-8-72} = \frac{x_3}{84-4}$$

$$\frac{x_1}{-40} = \frac{x_2}{-80} = \frac{x_3}{80} \Rightarrow \frac{x_1}{-1} = \frac{x_2}{-2} = \frac{x_3}{2}$$

$$X_3 = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$

Clearly, the eigen vectors X_1, X_2, X_3 are pairwise orthogonal, Since $X_1^T X_2 = 0, X_2^T X_3 = 0$ and $X_1^T X_3 = 0$.

Let N be the matrix whose columns are normalized eigen vectors.

∴ The normalized modal matrix is

$$N = \begin{pmatrix} 4/3 & 4/3 & -1/3 \\ -2/3 & 1/3 & -2/3 \\ -1/3 & 2/3 & 2/3 \end{pmatrix}$$

The transformation $X = NY$ reduces the given Q.F $Q = X^T A X$ into

$$Q = (NY)^T A (NY)$$

$$= Y^T N^T A N Y$$

$$= Y^T N^T A N Y$$

$$= Y^T (N^T A N) Y$$

$$= Y^T (D) Y \text{ where } D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -9 \end{pmatrix}$$

$$\text{i.e. } Q = (y_1 y_2 y_3) \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -9 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$\Rightarrow Q = 3y_1^2 + 6y_2^2 - 9y_3^2$, which is the required canonical form.

Hence the Q.F is indefinite in nature, as the canonical form contains both positive and negative terms.

Example : 2 Reduce the Q.F $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ into canonical form by an orthogonal reduction. Find its nature, rank, index and signature.

Solution : The given Q.F can be rewritten in matrix form as $A =$

$$\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

To find the eigen values

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$, where

S_1 = sum of the principal diagonal elements

$$= 6 + 3 + 3 = 12$$

S_2 = sum of the minors of principal diagonal elements

$$= \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix}$$

$$= (9 - 1) + (18 - 4) + (18 - 4) = 8 + 14 + 14 = 36$$

$S_3 = |A|$

$$= \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix}$$

$$= 6(9 - 1) - (-2)(-6 + 2) + 2(2 - 6)$$

$$= 48 - 8 - 8 = 32$$

$$\therefore \text{The characteristic equation is } \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

Solving by synthetic division method, we get

$$\begin{array}{r|rrrr} 2 & 1 & -12 & 36 & -32 \\ & & 2 & -20 & 32 \\ \hline & 1 & -10 & 16 & 0 \end{array}$$

$$\text{i.e., } \lambda = 2, \lambda^2 - 10\lambda + 16 = 0$$

$$\text{i.e., } \lambda = 2, (\lambda - 2)(\lambda - 8) = 0$$

$\therefore$ The eigenvalues are $\lambda = 8, 2, 2$

To find the eigenvector

$$\text{Let } (A - \lambda I)X = 0, \text{ where } X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{pmatrix} 6 - \lambda & -2 & 2 \\ -2 & 3 - \lambda & -1 \\ 2 & -1 & 3 - \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{aligned} (6 - \lambda)x_1 - 2x_2 + 2x_3 &= 0 \\ -2x_1 + (3 - \lambda)x_2 - x_3 &= 0 \\ 2x_1 - x_2 + (3 - \lambda)x_3 &= 0 \end{aligned} \right\} \text{--- (1)}$$

Case (i) : $\lambda = 8$

(1) $\Rightarrow$ becomes, $-2x_1 - 2x_2 + 2x_3 = 0$

$$-2x_1 - 5x_2 - x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -2 & 2 & -2 \\ \swarrow & \searrow & \swarrow \\ -5 & -1 & -2 \\ \swarrow & \searrow & \swarrow \\ -2 & -5 & -1 \end{array}$$

$$\frac{x_1}{2 - (-10)} = \frac{x_2}{-4 - 2} = \frac{x_3}{10 - 4}$$

$$\frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1}$$

$$X_1 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Case (ii): $\lambda = 2$

(1) $\Rightarrow$ becomes, $4x_1 - 2x_2 + 2x_3 = 0$

$$-2x_1 + x_2 - x_3 = 0$$

$$2x_1 - x_2 + x_3 = 0$$

All three equations are identical, we choose one equation. i.e. $2x_1 - x_2 + x_3 = 0$

Let $x_1 = 0$

$$\Rightarrow -x_2 + x_3 = 0 \Rightarrow x_2 = x_3 \Rightarrow \frac{x_2}{1} = \frac{x_3}{1}$$

$$X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Case (iii) : The third eigen vector is given by pairwise orthogonal conditions

$$X_1^T X_3 = 0 ; X_2^T X_3 = 0$$

$$\text{Let } X_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\therefore X_1^T X_3 = 0 \Rightarrow (2 \quad -1 \quad 1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow 2x_1 - 1x_2 + 1x_3 = 0$$

$$\&X_2^T X_3 = 0 \Rightarrow (0 \quad 1 \quad 1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow 0x_1 + 1x_2 + 1x_3 = 0$$

Solving, by the rule of cross-multiplication, we have

$$\begin{array}{ccc} -1 & 1 & 2 \\ \swarrow & \searrow & \swarrow \\ 1 & 1 & 0 \\ \swarrow & \searrow & \swarrow \\ 1 & 0 & 1 \end{array}$$

$$\frac{x_1}{-1 - 1} = \frac{x_2}{0 - 2} = \frac{x_3}{2 - 0}$$

$$\frac{x_1}{-2} = \frac{x_2}{-2} = \frac{x_3}{2} \Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{-1}$$

$$X_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

The normalized modal matrix N is

$$N = \begin{pmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{pmatrix}$$

The transformation $X = NY$ reduces the given Q.F $Q = X^T A X$ into

$$\begin{aligned} Q &= (NY)^T A (NY) \\ &= Y^T N^T A N Y \\ &= Y^T N^T A N Y \\ &= Y^T (N^T A N) Y \end{aligned}$$

$$= Y^T(D)Y \text{ where } D = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\text{i.e. } Q = (y_1 y_2 y_3) \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$\Rightarrow Q = 8y_1^2 + 2y_2^2 + 2y_3^2$, which is the required canonical form.

Hence the Q.F is positive definite in nature, as the canonical form contains only positive terms.

Rank of the Q.F is 3 (Since, the canonical form contains 3 square terms)

Index of the Q.F is 3 (Since, the number of positive square terms is 3)

Signature of the Q.F is 3 (Since, difference between positive and negative square terms is 3. i.e. $3 - 0 = 3$)

Problems for Practise

1. Reduce the quadratic form $2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3$ to canonical form by orthogonal transformation. Also find the rank, index, signature and nature of the Q.F.
2. Reduce the quadratic form $2x_1^2 + 5x_2^2 + 3x_3^2 + 4x_1x_2$ to canonical form by orthogonal transformation. Also find the rank, index and signature the Q.F.
3. Reduce the quadratic form $3x_1^2 - 3x_2^2 - 5x_3^2 - 2x_1x_2 - 6x_2x_3 - 6x_3x_1$ to canonical form by orthogonal transformation. Also find the rank, index and signature the Q.F.
4. Reduce the quadratic form $10x_1^2 + 2x_2^2 + 5x_3^2 - 4x_1x_2 + 6x_2x_3 - 10x_3x_1$ to a canonical form by orthogonal reduction. Find a set of values of x_1, x_2, x_3 which will make the form vanish. Examine its nature.

Answers

$$1. N = \begin{pmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \end{pmatrix}; Q = -y_1^2 + y_2^2 + 4y_3^2;$$

Rank=3; Index=2; Signature=1; Nature-indefinite

$$2. N = \begin{pmatrix} \frac{1}{\sqrt{5}} & 0 & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & 0 & \frac{1}{\sqrt{5}} \\ 0 & 1 & 0 \end{pmatrix}; Q = y_1^2 + 3y_2^2 + 6y_3^2;$$

Rank=3; Index=3; Signature=3

$$3. N = \begin{pmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{35}} & \frac{1}{\sqrt{14}} \\ 0 & \frac{-5}{\sqrt{35}} & \frac{2}{\sqrt{14}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{35}} & \frac{3}{\sqrt{14}} \end{pmatrix}; Q = 4y_1^2 - y_2^2 - 8y_3^2;$$

Rank=3; Index=1; Signature=-1

$$4. N = \begin{pmatrix} \frac{1}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{-3}{\sqrt{14}} \\ \frac{-5}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{14}} \\ \frac{4}{\sqrt{42}} & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{14}} \end{pmatrix}; Q = 3y_2^2 + 14y_3^2;$$

Positive semi-definite. $x_1 = 1, x_2 = -5, x_3 = 4$